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 Studierichting: Informatica
 Oefeningenreeks 3F nr. 1.4

$$f(x) = \frac{x^3 + 5x^2 - 7}{x^2 + 4x - 5}$$

$$\text{VA: } D = 4^2 - 4 \cdot (-5) = 36 \Rightarrow x_{1,2} = \frac{-4 \pm 6}{2}$$

$$x_1 = 1, x_2 = -5$$

$$\lim_{x \rightarrow 1+} f(x) = -\infty, \lim_{x \rightarrow 1-} f(x) = +\infty$$

$$\lim_{x \rightarrow -5+} f(x) = +\infty, \lim_{x \rightarrow -5-} f(x) = -\infty$$

$$\text{HA: } \lim_{x \rightarrow \infty} \frac{x^3}{x^2} = \infty$$

$$\text{SA: } a = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{x^3 + 5x^2 - 7}{x^3 + 4x^2 - 5x} = \lim_{x \rightarrow \infty} \frac{x^3}{x^3} = 1$$

$$b = \lim_{x \rightarrow \infty} \left(\frac{x^3 + 5x^2 - 7}{x^2 + 4x - 5} - x \right)$$

$$\lim_{x \rightarrow \infty} \frac{x^2 + 5x - 7}{x^2 + 4x - 5} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2} = 1$$

$$\Rightarrow ax + b = x + 1$$

$$\text{Ligging: } \lim_{x \rightarrow \infty} \left(\frac{x^3 + 5x^2 - 7}{x^2 + 4x - 5} - (x + 1) \right)$$

$$\lim_{x \rightarrow \infty} \left(\frac{x - 2}{x^2 + 4x - 5} \right) = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

x	-5	1	2
$f(x) - A$	$-$	$+$	$-$
			0
			$+$

$$x \rightarrow +\infty \Rightarrow f(x) \text{ boven } A$$

$$x \rightarrow -\infty \Rightarrow f(x) \text{ onder } A$$

References